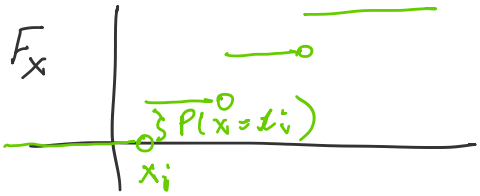
Lecture 7

Saturday, 5 September 2026 11:24 AM

Discrete RVs. → Support Countable

PMF → $P_X(k) = P(X=k)$, $k \in S_k$

$P(X \le x)$ ← CDF F_X



Eg Election with 2 candidates A & B.

60% supports A, 40% supports B.

Poll sample subset of population. (subset size → n)

X → # that support A

Poll chooses majority candidate as predicted winner

(a) $n = 5$, each selected ind. supports candidate A with prob. $p = .6$

P_w(poll incorrectly predicts) $= P(X \le 2) = F_X(2)$

$X = \sum_{i=1}^{n} \mathbb{1}_{\{i^{th} \text{ ind. supports A}\}}$

$X \sim Bin(n, p)$ $\quad n=5,\ p=.6$

$$P(X \le 2) = \sum_{k=0}^{2} P(X=k) = \sum_{k=0}^{2} \binom{n}{k} p^k (1-p)^{n-k}$$

(b) $n = 21$ → $P(X \le 10) = \sum_{k=0}^{10} \binom{21}{k} (.6)^k (1-.6)^{21-k}$

$\approx .3$

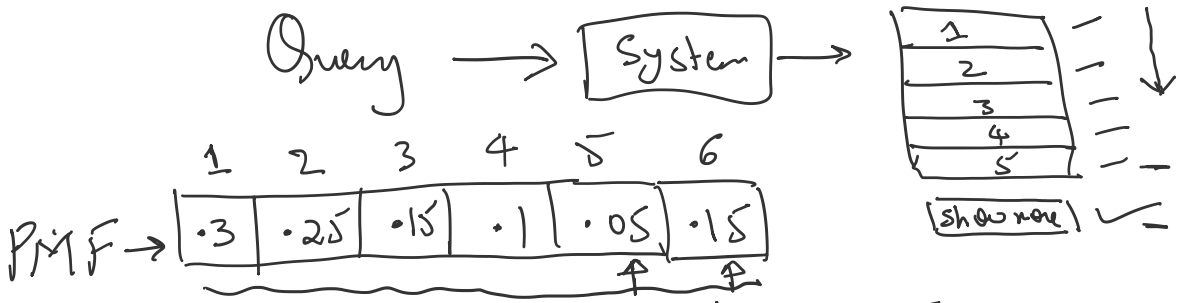
$\approx .25$??

(c) $n = 101$, $P(X \le 50) \approx .03$

Bin $X = \sum_{i=1}^{n} X_i$ $\quad X_i \sim Ber(p)$

X_i: 1 (p), 0 (1−p)

Eg Recommendation System.



X = position of user click ✓

$P(X \le 3) = F_X(3) = \sum_{i=1}^{3} P_X(i)$

$= .3 + .25 + .15 = .7$

$P(X > 4) = P(X=5) + P(X=6)$

$= .05 + .15 = .2$ ✓

$= 1 - P(X \le 4) = 1 - F_X(4)$

Continuous Random Variables

A rv is said to be continuous if there exists a non-neg fn $f_X : \mathbb{R} \to [0, \infty)$ s.t ($\exists$)

$$P(X \le x) = F_X(x) = \int_{-\infty}^{x} f_X(z)\, dz, \quad \text{for all } x \in \mathbb{R}$$

F_X : Prob. density fn (pdf)

(1) F_X is continuous and also differentiable (almost everywhere)



(2) $$\lim_{x \to \infty} F_X(x) = \lim_{x \to \infty} P(X \le x) = 1$$

$$\int_{-\infty}^{\infty} f_X(z)\, dz = 1$$

(3) $$P(a < X \le b) = F_X(b) - F_X(a)$$

$$= \int_{-\infty}^{b} f_X(z)\, dz - \int_{-\infty}^{a} f_X(z)\, dz$$

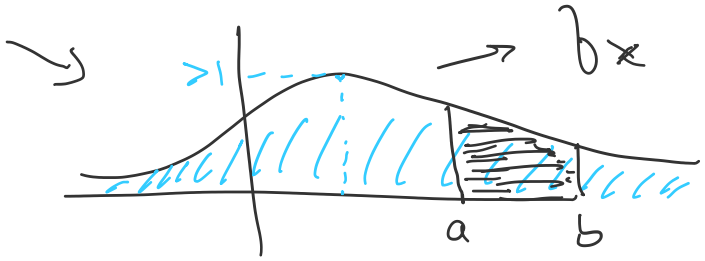
$$= \int_{a}^{b} f_X(z)\, dz$$

(4) $$P(X = a) = 0$$

$$\lim_{\delta \to 0} P(a-\delta < X < a) = 0$$

$$P(a-\delta < X \le a) = \int_{a-\delta}^{a} f_X(z)\, dz \approx \delta \cdot f_X(a) \to 0$$

$\lim_{x \to a}$

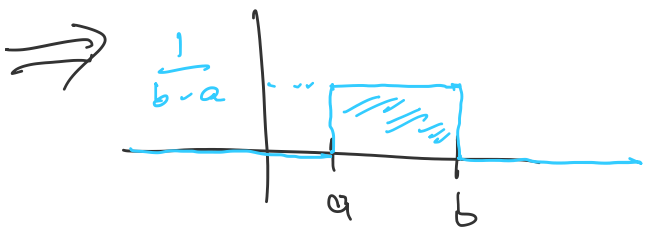


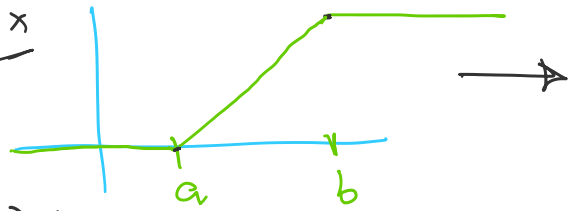
$$P(a \leq X \leq b) = P(a < X \leq b) + P(X = a)$$

$$= \int_a^b f_X(z)\,dz \qquad P(X=a) = 0$$

Eg ⓐ Uniform $[a, b]$ $\quad U([a,b])$

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{o.w.} \end{cases}$$





$P(X \leq x)$

$$F_X(x) = \int_{-\infty}^{x} f_X(z)\,dz$$

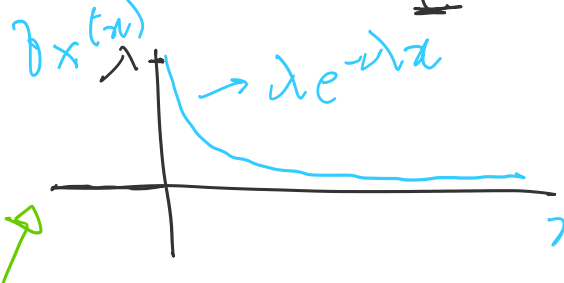
$P(X \leq x)$
$P(X \leq b)$

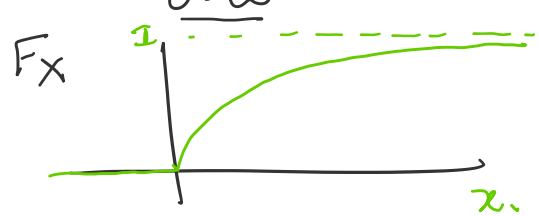
$$x \in (a,b) = \int_a^x \frac{1}{b-a}\,dz = \frac{x-a}{b-a}$$

Eg Exponential RV $Exp(\lambda)$

$\lambda > 0$

$$\rightarrow f_X(x) = \begin{cases} \lambda e^{-\lambda x} & , x \geq 0 \\ 0 & \end{cases}$$





$$F_X(x) = \int_{-\infty}^{x} f_X(z)\,dz = \int_{0}^{x} \lambda e^{-\lambda z}\,dz$$

$$= 1 - e^{-\lambda x}, \quad x \geq 0$$

Daily spend $\sim Exp(\lambda)$

$$\rightarrow P(X > 2000) = 1 - F_X(2000)$$

$$= e^{-\lambda \cdot 2000}$$

Memoryless Property → Exponential

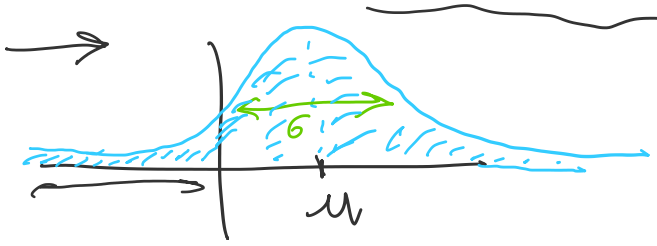
$$P(X > s+t \mid X > t) = P(X > s)$$

Eg → Gaussian dist. / Normal dist.

$X \sim N(\mu, \sigma^2)$

$\mu \in \mathbb{R}$ $\quad \sigma \in \mathbb{R}^+, > 0$

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$$



$x_1, x_2 \dots x_n$

$\bar{x}$

$h_i - \bar{x} \sim N$

CDF $\rightarrow$ $\Phi(x)$ $\rightarrow$ No closed form

i^{th} day $\xrightarrow[\text{pt.}]{}$ $y_i = a \cdot x_i + \varepsilon_i \sim N(0, \sigma^2)$

$Exp(\lambda) \rightarrow F_X(x) = 1 - e^{-\lambda x}, \quad x \geq 0$

$P(X \leq x)$

$P(X > x) = e^{-\lambda x}, \quad x \geq 0$

$P(X > s+t \mid X > t), \quad s, t \geq 0$

$= \dfrac{P(X > s+t \cap X > t)}{P(X > t)} = \dfrac{P(X > s+t)}{P(X > t)}$

$= \dfrac{e^{-\lambda(s+t)}}{e^{-\lambda t}} = e^{-\lambda s} = P(X > s)$